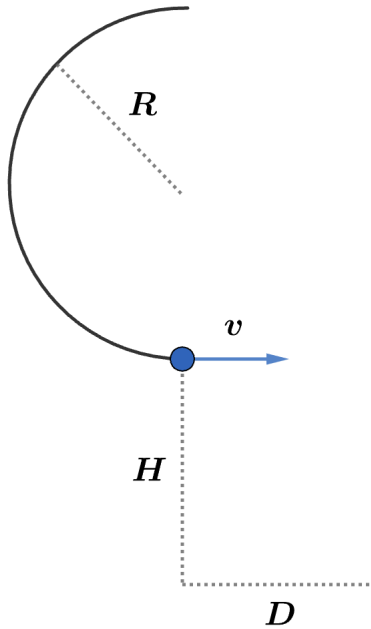


2016 F=ma Exam: Problem 19

Kevin S. Huang



By conservation of energy, when the bead travels to the bottom of the semicircle, it has speed

$$mg(2R) = \frac{1}{2}mv^2$$
$$v = 2\sqrt{gR}$$

From kinematics, it takes time

$$H = \frac{1}{2}gt^2$$
$$t = \sqrt{\frac{2H}{g}}$$

to fall a distance H . The range D is given by

$$D = vt = 2\sqrt{gR}\sqrt{\frac{2H}{g}} \propto \sqrt{RH}$$
$$D^2 \propto RH$$

Since D^2 and RH are proportional to each other, their graph is a line through the origin. Thus, the answer is D.